Online active testing

Anonymous authors
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Abstract

The abstract paragraph should be indented 1/2 inch on both left and right-hand margins. Use 10 point type, with a vertical spacing of 11 points. The word **Abstract** must be centered, in bold, and in point size 12. Two line spaces precede the abstract. The abstract must be limited to one paragraph.

1 Introduction

While unlabelled data are typically abundant, obtaining labels can be costly. In many scientific fields, assigning just one label may require specialised expertise and weeks of laboratory work or computation Bispo et al. (2025). Moreover, some labels provide more value than others, suggesting that one should strategically choose which examples to label to make the most of a limited budget. For instance, for autonomous driving, the majority of the data is redundant; thus, labelling most of it does not come with a performance improvement.

Depending on how the data are provided, two main scenarios can be considered: (1) In the *pool-based* setting, the learner has access to a large collection of unlabelled samples and can choose which ones to label in order to accomplish the task. (2) In the *online* setting (or *stream-based* setting), data arrive sequentially, one example at a time, and the learner must immediately decide whether to query its label or discard it. The data are never reconsidered.

Although this online setup offers less flexibility in selecting which points to label, it avoids the scalability issues associated with storing and handling a large pool of data. Inversely, it requires making immediate labelling decisions as data arrive, since instances cannot be stored and ranked like in pool-based settings. This constraint resembles the classic secretary problem Freeman (1983), where decisions must be made sequentially without seeing all options. Yet, considering this online setting is essential in real-world scenarios that demand real-time sampling, such as manufacturing quality control, video streaming, clinical trials, text classification, fraud detection, customer service interactions, and marketing, where the timely selection of new examples for labelling is critical for maintaining accurate and adaptive models Cacciarelli & Kulahci (2024).

This insight has driven substantial work on active selection of training labels Atlas et al. (1989); Settles (2009). Yet the expense of labelling test data has received far less attention Lowell et al. (2019). In controlled research environments, this is often overlooked because we can "cheat" by relying on very large test sets when evaluating sample-efficient training methods. For real-world practitioners, however, this becomes a significant problem: models must still be evaluated to select the best one and to build confidence in their predictions. For instance, Bispo et al. (2025) proposed an active learning sampling based on the model's uncertainty for machine learning potential and used the same sampling strategy to acquire the test set, which leads to a biased estimate of the overall generalisation capacities of the learned model.

When labelling is costly enough to require careful selection of training data, we likewise cannot afford to waste resources on large, fully labelled test sets. In fact, according to Lüth et al. (2023), evaluation is the main pitfall of active learning for real-world applications. Moreover, to emphasise the need for realistic evaluation methods, some studies have shown that active learning can even decrease classification performance in certain settings, a phenomenon referred to as the "cold start problem" Bengar et al. (2021); Gao et al. (2020); Mittal et al. (2019).

To tackle this challenge, the *active testing* framework for actively choosing which test examples to label has been proposed in the literature for both model selection and performance estimation. However, while the

pool-based active testing setting (all unlabelled data are accessible) has been largely investigated Sawade et al. (2010); Kossen et al. (2021; 2022); Mukherjee et al. (2022); Karimi et al. (2021), the *online* (or *stream-based*) active testing setting remains unexplored. In fact, most of the works in the stream-based setting focused on active model selection Karimi et al. (2021); Liu et al. (2022); Kassraie et al. (2023). Therefore, in this work, we develop acquisition functions that select test points on-the-fly so as to maximise the accuracy of the empirical risk estimate.

A core challenge in active testing is that selecting test points using information from the training data or the model under evaluation introduces sample-selection bias MacKay (1992); Dasgupta & Hsu (2008). For instance, querying points where the model is most uncertain Houlsby et al. (2011) tends to inflate the estimated test loss, since these points are usually harder than average. This effect is amplified for overconfident models, compromising our ability to compare models or tune hyperparameters. The fact that this correction is relatively new may explain why the active learning literature has largely overlooked active test-set selection: the bias is far more damaging for evaluation than for training Farquhar et al. (2021).

Another key factor is how the data themselves are generated. The instance–label pairs may be drawn i.i.d. from a fixed distribution, or they may be chosen in a more adversarial manner. While the i.i.d. assumption is sometimes reasonable, many real-world situations violate it, for example, when data exhibit temporal or spatial dependencies or when the underlying distribution shifts over time. In such cases, it is often more prudent to avoid strong assumptions and instead analyse performance under worst-case data streams.

1.1 Main Contributions

HS: In this work blablabla... + COntributions

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

2 Related Work

2.1 Active testing

Active Bayesian Assessment of Black-Box Classifiers qqs slides: `https://docs.google.com/presentation/d/19S9L_OCtgNHiYPrCoG2eNt_qvG2ghmHw20a4NvF6f44/edit?slide=id.g36f1df3e64d_0_0#slide=id.g36f1df3e64d_0_0`

SAWADE (self-normalised importance sampling): 2010: Active risk estimation
2010: Active Estimation of F-Measures
2012: Active Comparison of Prediction Models
2013: . Active evaluation of predictive models
Stratied sampling: 2010: Online stratified sampling: evaluating classifiers at web-scale
2011: Toward interactive training and evaluation
2012: Active evaluation of classifiers on large datasets
2018: Classifier Risk Estimation Under Limited Labeling Resources
2021: Active bayesian assessment of black-box classifiers
Kossen: Importance sampling 2021: Active Testing: Sample–Efficient Model Evaluation
2022: Active Surrogate Estimators: An Active Learning Approach to Label-Efficient Model Evaluation
Other: 2023: Actively Testing Your Model While It Learns: Realizing Label-Efficient Learning in Practice
2018: Active testing: An efficient and robust framework for estimating accuracy
2025: Risk Estimation with Active Labeling
2021: Sample efficient model evaluation
2024:All models are wrong, some are useful: Model Selection with Limited Labels
2025: Consensus-Driven Active Model Selection

Table 1: Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	**Lecture Notes**	14 point, bold
1st-level heading	**1 Introduction**	12 point, bold
2nd-level heading	**2.1 Printing Area**	10 point, bold
3rd-level heading	**Run-in Heading in Bold.** Text follows	10 point, bold
4th-level heading	*Lowest Level Heading.* Text follows	10 point, italic

2022:Chernoff Sampling for Active Testing and Extension to Active Regression

2.2 Unbiased active learning

Active learning for misspecified generalized linear models. 2007 Active learning in approximately linear regression based on conditional expectation of generalization error. 2006

Importance weighted active learning. 2009
Unbiased online active learning in data streams. 2011
Upal: Unbiased pool based active learning. 2012
Optimal sampling in unbiased active learning. 2020
On Statistical Bias In Active Learning: How and When To Fix It 2021

2.3 Stream/Online active learning

Active learning for data streams: a survey (2024)

Online active learning of decision trees with evidential data Online active learning ensemble framework for drifted data streams Drifting: https://sci-hub.ru/10.1109/TNNLS.2021.3091681 Online choice of active learning algorithms Second-order online active learning and its applications Fast kernel classifiers with online and active learning Online passive-aggressive active learning Active learning from imbalanced data: A solution of online weighted extreme learning machine

2.4 Stream sampling

`http://www.cohenwang.com/edith/Papers/varopt-full.pdf#page=1.28` Learn more, sample less: control of volume and variance in network measurement `https://dl.acm.org/doi/abs/10.1145/1066157.1066159` Truly Perfect Samplers for Data Streams and Sliding Windows

3 Online active testing setting

3.1 Active testing objective: estimation of the risk

Our goal is to estimate an evaluation statistic of a pretrained model $f_\theta : \mathcal{X} \to \mathcal{Y}$ in a sample-efficient manner. The pretrained model, f_θ, is fixed during the active testing procedure: we only evaluate it and do not update its parameters during the process. No assumptions about the form of the model are made: parametric or non-parametric, stochastic or deterministic. Moreover, the active testing method we propose applies to any supervised learning task (classification, regression, etc.). Depending on the problem, the evaluation statistic or metric may correspond to test accuracy, mean squared error, negative log-likelihood, or any other performance measure.

We assume an unknown data-generating distribution $p(x, y)$ over $\mathcal{X} \times \mathcal{Y}$. Evaluation statistics can be expressed as a risk over the test population. For generality, we express the risk as the expected loss under an arbitrary

loss function $\mathcal{L}$ as $\mathcal{R}(\theta) = \mathbb{E}_{(x,y)\sim p}[L(\theta; x, y)]$ *Our objective is to estimate* $\mathcal{R}(\theta)$ *accurately while querying as few labels as possible.*

3.2 Online active testing setting

In the online (or stream-based) active testing setting, we observe unlabelled samples according to their marginal, $x \sim p(x) = \int_y p(x, y)dy$, sequentially. At each time step, an agent can reveal a corresponding label $y \sim p(y \mid x)$ *only when we request it.* Formally, we consider a finite or infinite stream of samples $\{(x_t, y_t) \in \mathcal{X} \times \mathcal{Y}\}_{t\geq 1}$, where the labels y_t remain hidden unless queried.

To estimate $\mathcal{R}(\theta)$, we selectively query labels for a subset of the incoming samples. As a result, the distribution of labelled data generally differs from the marginal distribution $p(x)$. In particular, at each time step t, we define a *selection probability* $\beta_t = \beta_t(x_t)$ and draw a Bernoulli random variable $r_t \sim \mathcal{B}(\beta_t$ that determines whether the label is queried. If $r_t = 1$, we observe y_t and thus, the loss $\mathcal{L}(\theta; x_t, y_t)$, otherwise, both remain unobserved.

Remark: Noting that $p(r_t = 1 \mid x_t) = \beta_t$, the distribution of labelled samples, called the *instrumental distribution*, is $q_t(x_t, y_t) = p(x_t, y_t|r_t = 1)$. Similarly, unselected (unlabelled) samples follow $q_t^u(x_t, y_t) = p(x_t, y_t|r_t = 0)$. And finally, we can express the data-generating distribution, for any t, as: $p(x_t, y_t) = \beta_t q(x_t, y_t) + (1 - \beta_t)q_t^u(x_t, y_t)$.

The selection probability β_t can be adaptive, meaning it can depend on the previously selected (and unselected) samples. Formally, we require that $\beta_t \in \mathcal{F}_{t-1}$, where $\mathcal{F}_{t-1}$ denote the σ-algebra generated by all random variables up to time $t - 1$, i.e., by the observed points x_τ, their labeling decisions r_τ, and any observed y_τ for all $1 \leq \tau \leq t - 1$. Active testing in the sequential setting is more challenging than in the pool setting because the datapoints (x_t, y_t, r_t) are dependent on each other and unselected points are never reconsidered.

reservoir|batch

3.3 Batch setting

We also consider the batch setting where the unlabelled samples are also considered in a stream by batches of equal or unequal sizes. Formally, let's denote by N_t the size of batches at time t. The stream setting described in the previous paragraph is the limit case where $N_t = 1$ for all t. On the other hand, the pool-based setting is the limit where only one batch is observed.

3.4 Stopping criteria:

In online active testing (and online active learning), we acquire labels sequentially. We can choose to stop the acquisition, either:

1. after observing a total of T unlabelled samples x_t from the stream: under this criterion, the number of selected points n_T at time T is a random variable, but we construct the selection process according to a budget n_B on the sample size so that $\mathbb{E}[n_T] \leq n_B$ (practically, we can enforce the budget to be exactly met).

2. after collecting a total of n_B labelled samples: in this setting, the number of observed points T becomes a random variable. However, the budget n_B is always reached and never exceeded.

In the following sections, we propose "good" estimators of the risk $\mathcal{R}(\theta)$ *corresponding to each stopping criterion. Here, "good" refers to estimators that are unbiased or at least consistent, and that achieve variance reduction using adaptive importance sampling (AIS) principles.* The most naïve baseline will be called the iid baseline, which corresponds to selecting points to label according to a uniform distribution. The proposed estimates should have better properties than the naïve iid baseline, in particular with regard to the variance of the estimate.

4 Stopping criteria (1): Unbiased estimator:

We consider the setting where the stopping criterion is a number of samples seen through the selection process T. We denote n_T the number of labels collected. We assume that we are constrained by a budget n_B, so we have $\mathbb{E}[n_T] \leq n_B$. We can still fix a budget n_B corresponding to the maximum number of samples we can label. At time T, let us define the adaptive important sampling estimator:

$$\mathcal{R}_T(\theta) = \frac{1}{T}\sum_{t=1}^{T} \frac{p(x_t, y_t)}{p(x_t, y_t, r_t = 1)} r_t L(\theta; x_t, y_t).$$

As $p(x_t, y_t, r_t = 1) = \beta_t \, q_t(x_t, y_t)$. We have:

$$\mathcal{R}_T(\theta) = \frac{1}{T}\sum_{t=1}^{T} \frac{r_t}{\beta_t} L(\theta; x_t, y_t) = \frac{1}{T}\sum_{t=1}^{T} \frac{r_t}{\beta_t} \ell_t(\theta).$$

This estimator has already been used in pool-based active learning Beygelzimer et al. (2009); Ganti & Gray (2012); Imberg et al. (2020). In the pool-based setting, the exact budget can be matched as at every time step, one sample is labelled from the pool. In the online setting, the time steps correspond to the observation of unlabelled samples. Fortunately, we will see that the proposed estimator can eventually conserve good properties in the online setting in particular with regard to unbiasedness and the variance of the estimate.

Theorem 1. ***Unbiasedness*** *Let $T \geq 0$, if $\beta_t > 0$ for all $t \in [\![1, T]\!]$, then $\mathcal{R}_T(\theta)$ is an unbiased estimator of $\mathcal{R}(\theta)$: $\mathbb{E}[\mathcal{R}_T(\theta)] = \mathcal{R}(\theta)$*

Remark 1. *The importance distribution β_t does not have to be positive everywhere, for any t. It is enough to have $\beta_t > 0$ whenever $\ell_t(\theta) \neq 0$. For instance, if L is the 0-1 loss, only the uncorrected samples contribute to the estimate.*

Following the proof of Beygelzimer et al. (2009), we can adapt it to get the following result:

Theorem 2. ***Deviation and consistency:*** *Let consider the loss function L positive and bounded: $\forall x, y \sim p(x, y)$, $0 \leq L(\theta; x, y) \leq M$, with $M \in \mathbb{R}^{+*}$. Assume, there is a constant β_{min} such that $\beta_t \geq \beta_{min}$ for all $t \in [\![1, T]\!]$, then for any δ:*

$$P\left(|\mathcal{R}_T(\theta) - \mathcal{R}(\theta)| > \frac{M}{\beta_{min}}\sqrt{\frac{2\ln(2/\delta)}{T}}\right) < \delta \tag{1}$$

Remark 2. *Lire Langford & Schapire (2005) pour comprendre la différence avec la borne obtenu en supervision classique. Label complexity voir Beygelzimer et al. (2009).*

4.1 Analysis of the variance

In this section, we study the variance of our estimator $\mathcal{R}_T(\theta)$.

Theorem 3. ***Variance expression:***

$$V_{\mathcal{F}_T}[\mathcal{R}_T(\theta)] = \frac{V_{x,y}[L(\theta; x, y)]}{T} + \frac{1}{T^2}\sum_{t=1}^{T} \mathbb{E}_{x,y}\left[\frac{1 - \beta_t(x)}{\beta_t(x)} L(\theta; x, y)^2\right] \tag{2}$$

The first term is the variance of the Monte Carlo estimator if we observed all the labels. It is the best achievable variance for an estimator as it has access to all the information.

Remark 3. *This variance is to be compared to other simple sampling strategies, such as Monte Carlo in stream.*

Remark 4. *Assuming L bounded and $\beta \geq \beta_{min}$, we have:*

$$V[\mathcal{R}_T(\theta)] \leq \frac{V[L(\theta; x, y)]}{T} + \frac{1 - \beta_{min}}{\beta_{min}} M^2$$

Remark 5. *Let consider $\beta(x_t) = n/T$, then:*

$$V[\mathcal{R}_T(\theta)] = \frac{V[L(\theta; x, y)]}{n} + \frac{T - n}{nT} \mathcal{R}^2$$

The first term corresponds to the effective sample variance, and the second term is the importance weighting penalty, which reveals that the inverse-probability weighting increases variance when the mean loss is large.

4.2 Optimal sampling choice

Naive baseline: $\beta(x_t) = n/T$ As described in Imberg et al. (2020), one can chose the optimal sampling strategy in regard of the anticipated expectation defined by $\mathbb{E}_{y|x}[V[\mathcal{R}_T(\theta)|\mathcal{D}_T]]$.

Let us first consider the simple case where the sampling strategy is independent. In this case:

$$\begin{aligned} V[\mathcal{R}_T(\theta)|\mathcal{D}_T] &= \frac{1}{T^2} \sum_{t=1}^{T} \frac{(1 - \beta(x_t))}{\beta(x_t)} \ell_t(\theta)^2 \\ &= \frac{1}{T^2} \sum_{t=1}^{T} \frac{\ell_t(\theta)^2}{\beta(x_t)} - \frac{1}{T^2} \sum_{t=1}^{T} \ell_t(\theta)^2. \end{aligned}$$

Then, we have:

$$\mathbb{E}_{y|x}[V[\mathcal{R}_T(\theta)|\mathcal{D}_T]] = \frac{1}{T^2} \sum_{t=1}^{T} \frac{\mathbb{E}_{y|x}[\ell_t(\theta)^2]}{\beta(x_t)} - \frac{1}{T^2} \sum_{t=1}^{T} \mathbb{E}_{y|x}[\ell_t(\theta)^2]]. \tag{3}$$

The second term being a constant, only the first term remains to optimise.

Theorem 4. *If we fix an expected number of selected points (or a budget) such as $\sum_{t=1}^{T} \beta(x_t) = n$, we have:*

$$\beta^*(x_t) \propto n \sqrt{\mathbb{E}_{y|x}[\ell_t(\theta)^2]}. \tag{4}$$

Proof. We consider the following optimisation problem:

$$\begin{aligned} \min_{\beta} \quad & \mathbb{E}_{y|x}[V[\mathcal{R}_T(\theta)|\mathcal{D}_T]] \\ \text{s.t.} \quad & \sum_{t=1}^{T} \beta(x_t) = n. \end{aligned} \tag{5}$$

We have that:

$$\arg\min_{\beta} \mathbb{E}_{y|x}[V[\mathcal{R}_T(\theta)|\mathcal{D}_T]] = \arg\min_{\beta} \sum_{t=1}^{T} \frac{\mathbb{E}_{y|x}[\ell_t(\theta)^2]}{\beta(x_t)}$$

Using the method of Lagrange multipliers, we introduce the auxiliary function of the second term:

$$\Lambda(\beta, \lambda) = \sum_{t=1}^{T} \frac{\mathbb{E}_{y|x}[\ell_t(\theta)^2]}{\beta(x_t)} + \lambda \left(\sum_{t=1}^{T} \beta(x_t) - n \right)$$

Then, critical points of the Lagrangian are found by solving the equation system:

$$\Lambda(\beta, \lambda) = 0 \quad \Leftrightarrow \quad \begin{cases} \sum_{t=1}^{T} \beta(x_t) = n \\ \frac{\mathbb{E}_{y|x}[\ell_t(\theta)^2]}{\beta(x_t)^2} = \lambda \end{cases} . \tag{6}$$

Then, we get, $\forall t$, $\beta(x_t) = \sqrt{\frac{\mathbb{E}_{y|x}[\ell_t(\theta)^2]}{\lambda}}$. And by the first condition:

$$\sum_{t=1}^{T} \beta(x_t) = n \quad \Leftrightarrow \quad \beta(x_t) = \frac{n\sqrt{\mathbb{E}_{y|x}[\ell_t(\theta)^2]}}{\sum_{t'=1}^{T} \sqrt{\mathbb{E}_{y|x}[\ell_{t'}(\theta)^2]}}$$

□

Ok, now what if the sampling strategy is not independent? Actually it is the same as the non-diagonal terms do not depend of β. See proof above.

Remark 6. *TODO Zhan et al. (2022) https://arxiv.org/pdf/0803.0473 Keyword=stream sampling*

4.3 Consistency

Theorem 2.3 proves the consistency of the estimator. *********

4.4 Variance comparison and Label complexity

Let compare the anticapted variance of $\mathcal{R}_T(\theta)$ at β^* with the estimator obtained with uniform distribution $\beta_u(x) = n/T$ for any $t \leq T$. They are both unbiased, we can compare they variance:

$$\mathbb{E}_{y|x}[V[\mathcal{R}_T(\theta; \beta^*)|\mathcal{D}_T]] = \frac{1}{nT^2}\left(\sum_{t=1}^{T} \sqrt{\mathbb{E}_{y|x}[\ell_t(\theta)^2]}\right)^2 - \frac{1}{T^2}\sum_{t=1}^{T} \mathbb{E}_{y|x}[\ell_t(\theta)^2]]. \tag{7}$$

On the other hand:

$$\mathbb{E}_{y|x}[V[\mathcal{R}_T(\theta; \beta_u)|\mathcal{D}_T]] = \frac{1}{nT}\sum_{t=1}^{T} \mathbb{E}_{y|x}[\ell_t(\theta)^2] - \frac{1}{T^2}\sum_{t=1}^{T} \mathbb{E}_{y|x}[\ell_t(\theta)^2]]. \tag{8}$$

Using Cauchy-Schwartz inequality, we get that:

$$\mathbb{E}_{y|x}[V[\mathcal{R}_T(\theta; \beta^*)|\mathcal{D}_T]] \leq \mathbb{E}_{y|x}[V[\mathcal{R}_T(\theta; \beta_u)|\mathcal{D}_T]]. \tag{9}$$

Let compare the $\mathcal{R}_T(\theta)$ with $\mathcal{R}_t^{MC}$ for any $t \leq T$. They are both unbiased, we can compare they variance:

$$V[\mathcal{R}_T(\theta)] - V[\mathcal{R}_t^{MC}] = \frac{V[L(\theta; x, y)]}{T} - \frac{V[L(\theta; x, y)]}{t} + \frac{1}{T^2}\sum_{t=1}^{T} \mathbb{E}\left[\frac{(1-\beta(x_t))}{\beta(x_t)} L(\theta; x, y)^2\right]$$

Let consider $\beta(x_t)$ independent of $\mathcal{F}_{t-1}$ and all the x_t are iid. Then, we have:

$$V[\mathcal{R}_T(\theta)] - V[\mathcal{R}_t^{MC}] = \frac{V[L(\theta; x, y)]}{T} - \frac{V[L(\theta; x, y)]}{t} + \frac{1}{T}\mathbb{E}\left[\frac{(1-\beta(x))}{\beta(x)} L(\theta; x, y)^2\right]$$

https://arxiv.org/pdf/0812.4952 Compare optimal sampling with uniform sampling Voir yarin gal: active learning

4.5 Experiments:

prove unbiasedness and variance reduction. try various sampling strategies on toy exemples.

CIFAR

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A Appendix

You may include other additional sections here.